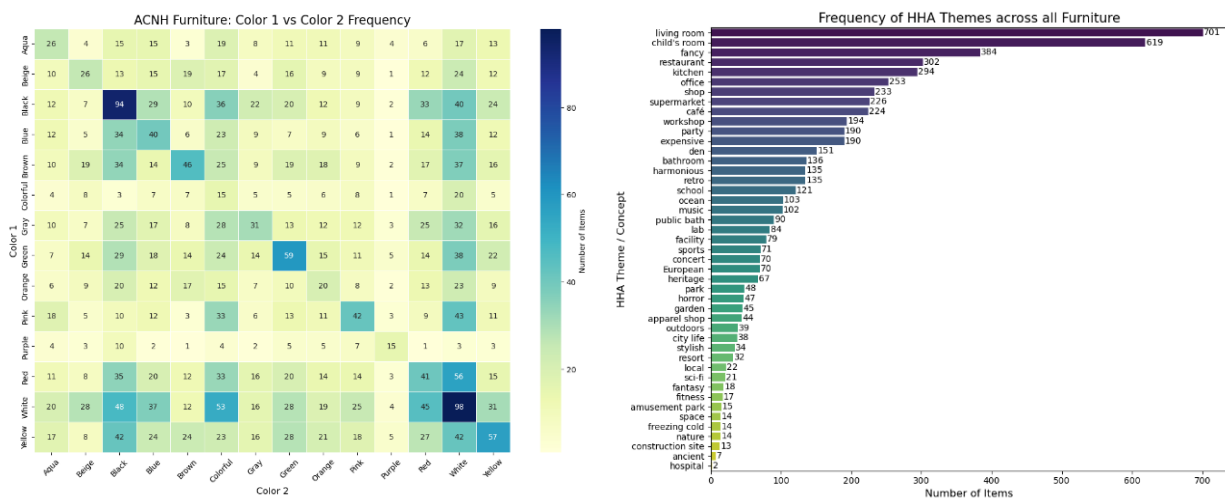


## Animal Crossing: New Horizons Recommender Systems

Animal Crossing: New Horizons is a free roam life simulation video game where players settle on an island and build a community with animal villagers through daily tasks, crafting items, and exchanging items. Animal Crossing: New Horizons offers a vast catalog of tens of thousands of decorative items such as furniture, decorations, insects, fish, and outfits that can be used to design a player's island. This project aims to recommend furniture items based on one query item. The recommended items will be similar in theme, color, and series.

The primary dataset this project will utilize is the Animal Crossing: New Horizons Data Spreadsheet, an extensive archive of all collectables, events, and villagers that is regularly updated and maintained by the ACNH community. The datasets are confined to only recommend items from the following sheets: housewares, wallpapers, floors, rugs, wall-mounted, ceiling décor, and miscellaneous items. These items can decorate both the island and indoor spaces. Although all variables are important to a player's agenda, the observed attributes are: Name to identify the item, variation, primary color (Color 1), secondary color (Color 2), craft-ability, primary HHA (Happy Home Academy) concept (HHA Concept 1), secondary HHA concept (HHA Concept 2), dimensions, where to obtain the item (source), type of item (Tag), buying price, selling price, and if the item is not purchasable (NFS). After merging the tables of each sheet and pruning variables, the data frame contained 14,341 items and 14 attributes.

To foreshadow any trends in the recommendations, a heatmap was curated of all the color combinations as well as an HHA theme frequency bar chart to show any biases.



In the heat map of colors, the most reoccurring colors are white and black. The combination White – White and Black – Black occurs the most. These colors may be recommended the most along

with a corresponding variation. In the bar chart of HHA themes, living room and child's room is the highest occurring theme in all items. This allows potential conflicts with distinguishing themes. With recommendation, a large bias towards one theme or a particular color combination can portray stronger similarity to the query item because that is the most dense and popular occurrence.

The four models in this project recommend similar items by calculating the distance between the query item vector and the feature vectors. Since the data is mostly categorical, non-numeric variables are turned into dummy variables. Binary values allow attributes to be treated as vector components. Numeric values such as buying price and selling price are combined to prevent redundancy and are log-scaled. Since many items have the same name but different variations, the data is reindexed as "item name (variation)". Two data frames were constructed: a weighted and unweighted data frame. The unweighted data frame treats all attributes equally whereas the weighted data frame prioritizes theme followed by series, tags, then colors. The corresponding weights are 5.0, 3.0, 2.0, and 1.0. The finalized data frames have 14,431 rows and 177 attributes. The first item in the database, the "? Block" from the Mario Bros games, was used as a test item in the first item but is found to be a degenerative case for later models since the tags and series do not align with the other Mario series items (Super Mushroom and Goal Pole). The other two items used for testing were "acoustic guitar (Cherry)" and "garden gnome (Reliable gnome)".

The first K Nearest Neighbors model utilizes a similarity matrix computed by cosine similarity. The model accepts a data frame, the query item, a similarity matrix, and the number of recommendation items. The query item is found in the data frame and the similarities to the query item vector are sorted in descending order. For the "? Block (None)", the unweighted data curated recommendations that were similar in color: the lily-pad table (yellow), baby bear (checkered), cute yellow wall (none), yellow stripes rug (none), and yellow quilt wall (none). The weighted data curated recommendations that are similar based on the weighted attributes: cool chair (white), garden table (green), floor monitor (none), bistro table (black), and floor seat (natural). In this model, the unweighted data finds smaller distances ranging from 0.7698 to 0.7709. The weighted data ranges from 0.8156 to 0.8207. The unweighted data using cosine similarity has a more cohesive aesthetic for the recommendations.

*Unweighted data with cosine recommender model, Query*

*Item: ? Block (None)*

| Furniture Name            | Score  |
|---------------------------|--------|
| lily-pad table (Yellow)   | 0.7709 |
| Baby bear (Checkered)     | 0.7705 |
| cute yellow wall (None)   | 0.7704 |
| yellow stripes rug (None) | 0.7702 |
| yellow quilt wall (None)  | 0.7698 |

*Weighted data with cosine recommender model,*

*Query Item: ? Block (None)*

| Furniture Name       | Score  |
|----------------------|--------|
| cool chair (White)   | 0.8207 |
| garden table (Green) | 0.8171 |
| floor monitor (None) | 0.8171 |
| bistro table (Black) | 0.8164 |
| floor seat (Natural) | 0.8156 |

The next K Nearest Neighbors model computes similarity using Euclidean distance. This model finds the nearest neighbors with the scikit learning neighbors module. The “garden gnome (Reliable gnome)” was used as a query item in this item as well as the “acoustic guitar (Cherry)”. The unweighted data recommends items with a smaller range of distance (1.4142 to 1.4145 for garden gnome) whereas the weighted data has a wider range (1.4142 to 1.7321 for garden gnome). Since Euclidean distance is the mean straight-line distance, the weights will increase the distance between the query vector and feature vectors. The unweighted data recommended a garden gnome of a different variation and 4 items similar in color. The weighted data recommended three garden gnomes with different variations and two bunny garden decorations with two variations indicating similar color, series, and tags.

*Unweighted data with Euclidean recommender model, Query*

*Item: garden gnome (Reliable gnome)*

| Furniture Name                 | Distance |
|--------------------------------|----------|
| garden gnome (Sprightly gnome) | 1.4142   |
| elegant brown rug (None)       | 1.4143   |
| simple small dresser (Brown)   | 1.4143   |
| outdoor folding table (Blue)   | 1.4144   |
| study chair (Dark brown)       | 1.4145   |

*Weighted data with Euclidean recommender model,*

*Query Item: garden gnome (Reliable gnome)*

| Furniture Name                  | Distance |
|---------------------------------|----------|
| garden gnome (Sprightly gnome)  | 1.4142   |
| bunny garden decoration (Brown) | 1.4143   |
| garden gnome (Passionate gnome) | 1.7321   |
| garden gnome (Rebel gnome)      | 1.7321   |
| bunny garden decoration (Ivory) | 1.7321   |

The next model also uses the scikit learning neighbors module. This model implements the Minkowski cubic distance. Setting  $p = 3$  exaggerates differences and emphasizes similarities. The difference between the query vector and the feature vector is cubed making a small difference much larger and similarities even smaller. The results for both weighted and unweighted data are identical. The drawback of this model is that the same item is recommended multiple times with different variations. For the “acoustic guitar (Cherry)” all other guitar variations were recommended. This is due to Minkowski strictly searching for the results that are nearly identical after cubing the distance between vectors.

*Unweighted data with Minkowski ( $p=3$ ) recommender model,*

*Query Item: acoustic guitar (Cherry)*

| Query Item: acoustic guitar (Cherry) | Distance |
|--------------------------------------|----------|
| acoustic guitar (Natural)            | 1.5874   |
| acoustic guitar (Black)              | 2.6684   |
| acoustic guitar (White)              | 2.7144   |
| acoustic guitar (Blue)               | 2.7144   |
| acoustic guitar (Brown)              | 2.7144   |

*Weighted data with Minkowski ( $p=3$ ) recommender model,*

*Query Item: acoustic guitar (Cherry)*

| Query Item: acoustic guitar (Cherry) | Distance |
|--------------------------------------|----------|
| acoustic guitar (Natural)            | 1.5874   |
| acoustic guitar (Black)              | 2.6684   |
| acoustic guitar (White)              | 2.7144   |
| acoustic guitar (Blue)               | 2.7144   |
| acoustic guitar (Brown)              | 2.7144   |

The last model uses Jaccard similarity to find the K nearest neighbors. The Jaccard method measures the similarity divided by the total number of features. This model works best with unweighted data. Jaccard is optimal for categorical data because it finds attributes that match. The distances calculated for the data are identical for some of the recommended items because of the “tie-ranking” between items with the same number of shared attributes. In the case of the garden gnome (Reliable gnome), all five recommended items’ distances were 0.3333. These results indicate that the five items: the study chair

(dark brown), garden gnome (sprightly gnome), simple chair (brown), simple small dresser (brown), and low screen (dark brown) all have the same number of similarities with the garden gnome (reliable gnome) vector. All these items are similar in color, tag, and concepts. Although this model is ideal for categorical data, the data is too sparse to break ties.

| Unweighted data with Jaccard recommender model,<br>Query Item: acoustic guitar (Cherry) |          | Unweighted data with Jaccard recommender model,<br>Query Item: garden gnome (Reliable gnome) |          |
|-----------------------------------------------------------------------------------------|----------|----------------------------------------------------------------------------------------------|----------|
| Query Item: acoustic guitar (Cherry)                                                    |          | Query Item: garden gnome (Reliable gnome)                                                    |          |
| Furniture Name                                                                          | Distance | Furniture Name                                                                               | Distance |
| Bunny Day table (None)                                                                  | 0.3636   | study chair (Dark brown)                                                                     | 0.3333   |
| clackercart (Cool)                                                                      | 0.3636   | garden gnome (Sprightly gnome)                                                               | 0.3333   |
| acoustic guitar (Natural)                                                               | 0.3636   | simple chair (Brown)                                                                         | 0.3333   |
| artful screen (Black)                                                                   | 0.4545   | simple small dresser (Brown)                                                                 | 0.3333   |
| artful chair (Cherry wood)                                                              | 0.4545   | low screen (Dark brown)                                                                      | 0.3333   |

Since the results of these models are not predicting or fitting, conventional error analysis cannot be performed. Evaluating results of these recommendations is subjective since there is no user data to test if the recommended items are in fact desired or relevant. The closest theoretical estimation would be similarity in color, theme, or series. The performance of the models is shown in the ensemble model. Since all models measure the distance between similar feature vectors and the query vectors, the ensemble model normalizes all the data, runs each model, and weighs each of the scores produced. The following models are incorporated: weighted and unweighted cosine recommender, weighted and unweighted Euclidean recommender, Minkowski recommender, and the Jaccard recommender. Their corresponding weights are 0.4, 0.1, 0.3, 0.1, 0.05, and 0.05. One of the test items was the garden gnome (Reliable gnome).

| Ensemble recommender model,<br>Query Item: acoustic guitar (Cherry) |                            |                | Ensemble recommender model,<br>Query Item: garden gnome (Reliable gnome) |                                 |                |
|---------------------------------------------------------------------|----------------------------|----------------|--------------------------------------------------------------------------|---------------------------------|----------------|
|                                                                     | item                       | weighted_score |                                                                          | item                            | weighted_score |
| 4                                                                   | acoustic guitar (Natural)  | 0.500000       | 5                                                                        | garden gnome (Sprightly gnome)  | 0.431833       |
| 1                                                                   | acoustic guitar (Black)    | 0.466815       | 0                                                                        | bunny garden decoration (Brown) | 0.399943       |
| 2                                                                   | acoustic guitar (Blue)     | 0.350000       | 3                                                                        | garden gnome (Passionate gnome) | 0.350071       |
| 3                                                                   | acoustic guitar (Brown)    | 0.350000       | 4                                                                        | garden gnome (Rebel gnome)      | 0.350071       |
| 5                                                                   | acoustic guitar (White)    | 0.350000       | 1                                                                        | bunny garden decoration (Ivory) | 0.350000       |
| 6                                                                   | artful chair (Cherry wood) | 0.150000       | 8                                                                        | outdoor folding table (Blue)    | 0.136270       |

The results shows that the garden gnome (Sprightly gnome) was the top recommendation with a score of 0.431833 verifying that feature weighing is successfully accounted for. Following is the bunny garden decoration (Brown) with a score of 0.399943 and after that is a tie between garden gnome (Passionate gnome), garden gnome (Rebel gnome), and bunny garden decoration (Ivory). Lastly, is the outdoor folding table (Blue). The bunny garden decoration (Brown) being second over the tied tier of recommendations exhibits tag and theme priority. The tied recommendations show that the lower performing models, Jaccard and Euclidean, have difficulties breaking ties. Including more attributes may

remedy this issue. From these recommendations, the weighted cosine model is the primary model and Jaccard is the least prioritized to avoid far off recommendations.

If this project was not bound to the constraints of time, recommendations would have more nuances and intricacy. Although the model does recommend items that are relevant to the query items, an aspect that would be helpful is incorporating villager preferences. Players can then be recommended items that will enhance their relationships with the villagers. If time allowed, synthesizing user data would also be implemented to better evaluate models and to exercise more of the curriculum on the project objective. A recommender system for island items in Animal Crossing: New Horizons fosters creativity and a more personalized experience while making a sandbox open-world game continuously engaging and entertaining.